



Land and Site Valuation **19**

Land has value because it provides potential utility as the site for a structure, recreational facility, agricultural tract, right of way for transportation routes, water storage, or other use. Land may also have value when valuable minerals (e.g., oil, coal, gravel, sand, iron ore) can be extracted from it. In some locales, the ability to extract water from a site is very valuable, and the legal right for its use may dwarf all other forms of utility. In other areas, the ability to build in the air space above a parcel of land may have as much or more value than the potential uses of the surface or sub-surface of the land. In sum, if land has utility for a specific use and there is demand for that use, then the land has value to a particular category of users. Use characteristics and demand are not static and will vary over time and under changing economic conditions. Beyond the basic utility of land, however, many principles and factors must be considered in the process of land valuation.

Relation to Appraisal Principles

The appraisal principles of anticipation, change, supply and demand, substitution, and balance all influence land value. Anticipation means that value is created by the expectation of benefits to be derived in the future. For example, if buyers anticipate that land in a certain location will be in demand for office use within the next five years, they may be motivated to acquire land for future development even though the development of office space is not presently feasible. The competition among the buyers in the market for land creates a price level for the land that may have little to do with the current use.

In comparison to most commodities, the supply of land is relatively stable. Although vast changes have occurred in the earth's surface over the ages and slight modifications in the supply and quality of land may occur over time, these changes usually occur over such a long term that their effects are imperceptible in real estate markets. There are, however, a few notable exceptions to the permanence of land, such as the accretion or erosion of land along a shoreline, the pollution of land with

harmful wastes, the exhaustion of agricultural land through improper farming methods, and the transformation of arable land into arid land due to ecological imbalances or climate change. Earthquakes may change the surface of the earth, faults beneath the surface can create vast sinkholes, and old underground mines can cause subsidence.

The principle of substitution, which holds that a buyer will not pay more for one property than for an equivalent property, applies to raw land and developable sites just as it does to improved property. The principle of substitution indicates that the greatest demand will be generated for the lowest-priced site with similar utility.

The principle of balance is also applicable to the value of land. When the various elements of a particular economic mix or a specific environment are in a state of equilibrium, the demand for land is sustained. When the balance is upset, demand may decrease and values change. For example, if an industrial district has too much industrially zoned land and a declining number of industrial users, then the demand for industrial land will probably fall. On the other hand, if the industrial district is in transition to other, more profitable uses and there is a reasonable probability of a zoning change, the value of a particular site could be higher than the value for continued industrial use.

Property Rights and Public Controls

The analysis of the value of a parcel of land focuses on the physical parcel and the accompanying property rights. These rights may include the right to

- Develop the land to its highest and best use
- Lease the land to others
- Farm the land
- Mine the land
- Alter the land's topography
- Subdivide the land
- Assemble the parcel of land with other parcels
- Hold the land for future use
- Construct or alter building improvements
- Reserve the land in perpetuity via an easement, donation, or sale to a nonprofit or government entity

Whenever possible, appraisers consult title reports, public records, or available land survey information to identify easements, rights of way, and private or public restrictions that affect the subject property.

In an effort to encourage planned growth and compatibility among different land uses, governments regulate how land can be used. Most municipalities and counties have some form of land use regulations that specify how a parcel of land can be developed. In addition to land use regulation, many jurisdictions have master plans or comprehensive land use plans that specify long-term development goals. Frequently, real estate developers must have amenities such as open space, streets, and off-site public improvements in place or dedicated before a proposed development receives approval from the appropriate public agency. Development can proceed only after detailed development plans have been submitted and zoning and building department approvals have been obtained. In many areas, citizen groups will protest a development that they do not like, and their objections can influence the type of

development that is finally approved and the cost and time required to get approvals. In some jurisdictions, the time, effort, and costs to obtain approvals are significant. Typically, the value of the land increases after formal site plan approval has been achieved and permits have been obtained for a development. At that point, the site has full entitlements and is ready for development.

Through the power of eminent domain, the government can acquire land from the private sector to be used for public and sometimes non-public projects, to augment the supply of public land, and to encourage economic development or eliminate blight in a specific area.¹

In some jurisdictions, transferable development rights can greatly affect the value of land. In urban areas, development rights can be valuable, and discrete markets have formed in which these rights are traded. In some rural areas, government agencies compensate farmers for retaining land in agricultural use, and the agencies then shift or sell the benefit of those development rights to other locations. Lower ad valorem taxes on agricultural land also affect rural land use. This form of tax subsidy tends to extend the duration of agricultural uses.

A trend in the United States has been the acquisition of land through open space or conservation easements that are held in perpetuity by qualified agencies. These permanent encumbrances limit or prohibit the development potential of land. The potential uses of land subject to perpetual open space or conservation easements are usually restricted, as specified in the deed of easement, and thus the value of the land is affected. Open space can also be preserved through the qualified donation of specified easement rights to a qualified recipient such as a land trust.

Water, mineral, and air rights are also important considerations in land valuation. Water rights cover the flow of water, usually for stated times and in stated quantities, for irrigation and hydroelectric power generation. Water rights also can take the form of riparian rights which, under common law, grant a landowner the ownership of waters that share a border with the owner's parcel of land. Mineral rights cover the underground portion of the land and usually refer to the right to extract underground minerals or to use underground caverns or reefs for storage. The valuation of mineral rights usually requires specialized research, but appraisers need to understand if mineral rights are to be included or excluded from the valuation. Real estate developments established on air rights, often over railways, can be found in older urban areas, particularly where the density of the built environment is already high and little land is available for new construction.

Physical Characteristics and Site Improvements

In the common parlance of valuation, land becomes a *site* when the parcel of land is improved and ready to be used for a specific purpose. The physical characteristics of a site, the utilities available, and any site improvements affect the use and value of the land. The physical characteristics of a parcel of land that an appraiser must consider include

- Size
- Shape

1. The taking of private land through condemnation for economic development, and particularly for private sector development, has come under greater public scrutiny since 2005 when the US Supreme Court issued its decision in *Kelo v. City of New London*.

- Frontage
- Soils
- Location
- View
- Topographical characteristics such as contour, grade, and drainage

A site may have both on-site and off-site improvements that make it suitable for its intended use or for new construction. The availability of water, sewers, electricity, natural gas, and telephone and data lines also influences the use and development potential of a parcel of land. (See Chapter 12.) On-site improvements are subject to physical deterioration and functional obsolescence like buildings and other structures when the site is valued as improved. Off-site improvements may have an effect on the value of a site, even though they are not a part of the site.

Highest and Best Use

The valuation of land draws directly from the conclusions of highest and best use analysis. Even if a site is already improved, the site is valued as though vacant and available for development to its highest and best use. Consideration of the site as though vacant facilitates the orderly analysis and solution of appraisal problems that require land to be valued separately. The highest and best use of a competitive site on the date of sale is the basis of the comparability of that site to the property being appraised. Regardless of how physically similar a potentially comparable site is to the subject site, the most comparable sales would have the same or a similar highest and best use. Markets without recent sales of comparable sites are problematic and may require analysis of competing sites where sufficient data are available or alternative land valuation methods may be considered.

The highest and best use of the property as improved is affected by how much the existing improvements contribute to the value of the property as a whole. The value contribution of the improvements can be estimated by subtracting the market value of the site from the market value of the property as improved. Site value may be equal to or greater than the value of the property as improved even when substantial improvements are located on the site. Any penalty for the misuse of the site is applied to the improvements. Demolition of existing improvements is usually appropriate when those improvements do not contribute to the overall property value. In that situation, the cost of converting the property into vacant land is deducted from the value indication for the site.

Sometimes the highest and best use of a property is to keep the land vacant until land values rise to support new development. In that situation, the most appropriate comparable properties would be other vacant sites being held for future development for similar uses.

Excess and Surplus Land

As first discussed in Chapter 12, excess land is land that is not needed to support the improvements and may be sold off separately from the rest of the property at its own highest and best use. An area of excess land may have a different highest and best use from the rest of the site, which must be addressed in the highest and best use analysis by testing alternate uses of the excess land for physical possibility, legal permissibility, financial feasibility, and maximum productivity in the market. Further-

more, excess land has to be treated separately in the valuation process. An entirely different set of comparable data may be required to analyze the value influences on the excess land, and the appraiser must consider and report the contributory value of the excess land to the value of the whole. Adding the value indication of the excess land to the value indication of the rest of the property may or may not be appropriate because the sum of the parts may or may not equal the value of the whole.

In contrast, surplus land does not have a separate value from the rest of the site because it cannot be sold separately. It is extra land that may or may not contribute value to the overall property. Both surplus and excess land are extra meaning they are not needed to support the improvements. However, excess can be sold off with its own highest and best use whereas surplus land cannot. Surplus land may or may not have the same value per unit of comparison (e.g., value per square foot, value per acre) as the rest of the site.

Applicability and Limitations of Valuation Techniques

Sales comparison is usually the preferred method for developing an opinion of site value. When this method is used, most of the techniques for selecting comparable sales and making adjustments that are described in Chapter 20 can be applied to site valuation. When there are not enough sales of similar parcels for the application of sales comparison, alternative methods such as market extraction, allocation, land residual analysis, and various income capitalization techniques may be used.

Sales Comparison

Sales comparison may be used to value land that is actually vacant or land that is being considered as though vacant for valuation purposes. Sales comparison is the most common technique for valuing land, and it is the preferred method when comparable sales are available. To apply this method, data on sales of similar parcels of land is collected, analyzed, compared, and adjusted to provide a value indication for the site being appraised. In the comparison process, the similarity or dissimilarity of the parcels is considered.

Appraisers perform several tasks in developing an opinion of site value:

- Identify the highest and best use and other characteristics of each potential comparable sale and then choose the appropriate properties for analysis.
- Gather data on actual sales as well as listings, offers, and options.
- Identify the similarities and differences in the data.
- Identify units of comparison that explain market behavior.
- Adjust the appropriate unit prices of the comparable sales to account for the dissimilar characteristics of the land being appraised.
- Form a conclusion as to the value of the subject site.

The objective of sales comparison is to select the most comparable sales and then adjust the comparable sales for differences that cannot be eliminated within the selection process. Elements of comparison may include property rights, financing terms, conditions of sale (motivation), expenditures immediately after purchase, market conditions (changes over time), location, physical characteristics, economic characteristics, available utilities, and zoning. The physical characteristics of a parcel of land include, but are not limited to, its size, shape, frontage, topography, soil conditions, location, and

Table 19.1 Applicability and Limitations of Land Valuation Methods**Sales Comparison**

Procedure	Sales of similar, vacant parcels are analyzed, compared, and adjusted to provide a value indication for the land being appraised.
Applicability	Sales comparison is the most common technique for valuing sites, and it is the preferred method when comparable sales are available.
Limitations	A lack of sales and the comparability of the available data may weaken support for the value estimate.

Market Extraction

Procedure	An estimate of the contributory value of improvements is deducted from the total sale price of a property to arrive at an indicated land value for the comparable. The indicated land values of the comparables are then compared to provide a value indication for the land being appraised.
Applicability	This technique is most applicable when the contribution of the improvements to total property value is generally small and relatively easy to identify. (The technique is frequently used in rural areas.)
Limitations	The appraiser must be able to determine the value contribution of the improvements.

Allocation

Procedure	A ratio of site value to property value is extracted from comparable sales in competitive locations and applied to the value of the improved subject property or comparable properties to develop the site value.
Applicability	This technique is applicable when • Valuing one-unit residential lots where ample sales of improved properties are available and the ratio of site value to improved property value can be supported. This method tends to be less accurate for commercial properties, especially when the number of vacant land sales is inadequate. • For commercial properties or where relatively few sales are available, allocation can provide a test of reasonableness rather than a formal opinion of site value.
Limitations	The allocation method does not produce conclusive value indications unless ample sales data is available. The method is rarely used as the primary land valuation technique for properties other than residential subdivision lots. Also, land-to-property value ratios can be difficult to support.

Land Residual Analysis

Procedure	The net operating income attributable to the land is capitalized or the cost to construct an improvement is deducted from the value as if completed to produce an indication of the land's contribution to the total property.
Applicability	This technique is applicable in the financial analysis of alternative uses of a particular site in highest and best use analysis and when land sales are not available.
Limitations	When used as an income capitalization technique, the following conditions must be met: 1. Building value is known or can be accurately estimated. 2. Net operating income to the property is known or can be estimated. 3. Both building and land capitalization rates are available from the market. When using a cost-based technique, the appraiser must be able to determine the value contribution of the improvements, estimated at their depreciated cost

Ground Rent Capitalization

Procedure	A market-derived capitalization rate is applied to the ground rent of the subject property.
Applicability	This method is useful when comparable rents, rates, and factors can be developed from an analysis of sales of leased land or sales of unleased land for which the market rent can be credibly determined.
Limitations	An adjustment to the value indication for property rights may be necessary.

Subdivision Development Analysis (Discounted Cash Flow Analysis)

Procedure	Direct and indirect costs and entrepreneurial incentive are deducted from an estimate of the anticipated gross sales price of the finished lots or units, and the net sales proceeds are discounted to present value at a market-derived rate over the development and absorption period. If entrepreneurial incentive is not deducted as a line-item expense, then the discount rate must reflect the full effect of any profit.
Applicability	This technique is applicable when subdivision development is the highest and best use of the land and there is market support for immediate absorption.
Limitations	Discounted cash flow analysis requires significant amounts of data such as development costs, profit margins, sales projections, and the pricing of developed lots or units, together with a supportable forecast of market absorption.

Note: Certain US states do not recognize subdivision development analysis as an acceptable valuation method for purposes of litigation valuation.

view. (For a detailed discussion of elements of comparison, see Chapter 21.) Unit prices may be expressed as price per square foot, front foot, acre, lot, dwelling unit, floor area ratio (FAR), or other unit used in the market.

If sale prices have been changing rapidly over the past several years and an adequate amount of sales data is available, the sales selected for comparison should take place as close as possible to the effective appraisal date. When current data on local sales is not available, appraisers may need to expand the search to another market area, which may call for an adjustment for location, or extend the search back in time in the same market area, which usually calls for an adjustment for market conditions. The decision to use sales from other market areas or older sales should be based on which adjustment has more support—the location adjustment or the market conditions adjustment.

Among generally similar sales, size may be less important as an element of comparison than date and location. Most land uses have an optimal site size. If the site is too large, the value per unit of comparison for the surplus land tends to decline at an accelerating rate. Because sales of different sizes may have different unit prices, ordinarily more weight is given to sales of comparable properties that are approximately the same size as the subject property.

Zoning is a basic criterion in selecting comparable sites. Sites zoned the same as the subject property generally have the same or a similar highest and best use and may be the most appropriate comparable properties. However, zoning can be less important than utility or highest and best use in areas that are in transition or targeted for redevelopment. If sufficient sales in the same zoning category are not available, data from similar zoning categories can be used and adjustments may be necessary.

In addition to recorded sales and signed contracts, appraisers may consider offers to sell (listings) and offers to purchase. However, offers generally provide less reliable data than signed contracts and completed sales because they represent what a buyer was willing to pay rather than the price both a buyer and seller were willing to agree to consummate a sale. Also, appraisers may choose to use a recent sale of the subject property as a comparable sale. For example, if the subject site sold 18 months ago for \$545,000 in a market that is increasing at 3% per year (on a straight-line basis, i.e., not compounded), an appraiser could derive an indication of market value by adjusting the prior sale for the changes in the market:

$$\$545,000 \times [1 + (1.5 \times 0.03)] = \$569,525$$

This assumes that the property has not changed physically since the prior sale, the prior sale meets all the requirements of a market sale, and the prior sale occurred within a reasonable period from the effective date of value. It also assumes that, during the intervening period, the applicable zoning and land use regulations did not change.

Data on land sales is available from sources such as data services, online and print publications, and deed and assessment records. Interviews with the parties involved in transactions—i.e., the buyers, sellers, lawyers, and brokers—provide more direct information and may reveal adjustments that should be made for conditions of sale or sale concessions. The interviews should identify the buyer motivations and planned use as well as the status of any approvals and entitlements.

Sales comparison is the most commonly used method of valuing land. Data on sales of similar parcels of land is collected, analyzed, compared, and adjusted to reflect the similarity or dissimilarity of those parcels to the subject property.

After comparable data is collected and categorized, and the comparable properties are examined and described, sales data can be assembled in an organized, logical manner. Sales are commonly arrayed in a market data grid that identifies the elements of comparison that may require adjustments. Appropriately developed adjustments for significant differences between the subject property and the comparable properties may be made to the sale or unit prices of the comparable properties using a variety of techniques. (Techniques for making adjustments in sales comparison analysis are discussed in Chapter 21.)

Generally, separate adjustments are made to the comparable sales for each element of comparison. The magnitude of each adjustment is indicated by the data and the judgment of the appraiser. If the data selected is not sufficient to support the required adjustments, the appraiser should gather and analyze additional comparable data.

A sale price adjustment may be a precise dollar amount or percentage developed from market evidence. Either way, the adjustments are the appraiser's approximations of the market's reaction to the element of comparison. Adjustments can be totaled and factored into the comparable sale prices as part of the initial data gathering process. Typically, adjustments are made in a preferred order—i.e., transactional adjustments for property rights, financing, and sale and market conditions are made before property adjustments for location and physical characteristics. All adjustments should be presented in the appraisal report in a logical and understandable manner. If specific adjustments cannot be supported by market evidence, the comparative analysis of sale prices can still help appraisers analyze a comparable property's relative superiority or inferiority to the subject property. This fine tuning can take place in the reconciliation.

Market Extraction

Market extraction is a valuation technique in which land value is estimated by deducting the contributory value of the improvements from the sale price of an improved comparable property. The remainder represents an indication of the value of the land. Improved sales in rural areas are frequently analyzed in this way because the building and site improvements may contribute little value in comparison to the underlying land value. The improvement contribution is typically small and relatively easy to quantify.

As an example, consider a vacant subject site in an area where few sales of comparable sites have occurred recently. The sales summarized in Table 19.2 involve sites that are similar to the subject property except for the improvements. The contributory value of the improvements is subtracted from each sale price to calculate value indications for the sites of the comparable properties (Table 19.3). Those value indications can, in turn, be analyzed using sales comparison techniques and reconciled into a value indication for the subject site. In this case, the value indications range from \$407,000 to \$435,000.

Allocation

The allocation technique is based on the principle of balance and the related concept of contribution. Both affirm that there is a normal or typical ratio of land value to property value for specific categories of real estate in specific locations. Meaningful support for an allocation ratio may be derived from a variety of sources such as observed patterns over time in an area and consultation with developers who sell improved properties and can allocate sale prices between the land and the improvements based on their costs.

Table 19.2 Comparable Sales for Market Extraction

Sale	Sale Price	Description
1	\$450,000	Includes storage building that contributes \$15,000
2	\$465,000	Includes 1,500-sq.-ft. building with a contributory value of \$32 per square foot
3	\$432,000	Includes permits and entitlements for construction that contribute \$25,000 in value
4	\$448,000	Includes a temporary sales building that contributes \$32,000

Table 19.3 Market Extraction of Site Value

	Sale 1	Sale 2	Sale 3	Sale 4
Sale price	\$450,000	\$465,000	\$432,000	\$448,000
Less contribution of improvements	– \$15,000	– (1,500 × \$32)	– \$25,000	– \$32,000
Site value indication	\$435,000	\$417,000	\$407,000	\$416,000

In situations where there is limited sales data, the allocation technique can be used to establish approximate land value when the number of vacant land sales is inadequate. For example, an appraiser could use allocation to value the site for a new one-unit home in a large, newly developed subdivision where few sales of vacant land have occurred but credible data from several recent sales of improved properties is available. The sale prices of new homes in the development range from \$275,000 to \$315,000, and the developer reports that site values within the subdivision range from 15% to 20% of sale prices. Based on these figures, the indicated range of value for the sites would be from \$41,250 to \$63,000.

The most common application of allocation is in the analysis of sales of residential subdivision lots, where the appraiser can directly measure the ratio of site value to total property value. Allocation is rarely used as the primary method of site valuation for commercial properties because of the relatively large number of sales needed to support a credible value opinion when many adjustments for transactional and property differences are necessary. In addition, commercial properties may vary widely in parcel size and intensity of use. Parcels that have more land than is necessary for the existing improvements are also difficult to value using allocation.

The allocation technique, and sometimes the market extraction technique, can be difficult to use in markets where the highest and best use and land value ratios of comparable parcels are not similar to the subject property. For example, consider a subject site that is zoned for commercial use but is improved with a residence. The commercial land value is \$500,000 and the improved property value is \$550,000. In this case, if the comparable sites did not have similarly high ratios of land value to improvement value, the allocation and extraction techniques would give misleading indications of value.

Land Residual Analysis

Land residual analysis involves producing an indication of the land's contribution to the total property by either (a) capitalizing the net operating income attributable to the land or (b) deducting the cost to construct an improvement from the value of the subject property as if improved. By isolating the value contribution of the land, the land

residual technique serves as an essential tool in highest and best use analysis to test the economic productivity of alternate uses of the site as though vacant. (See Chapter 18.)

As an income capitalization technique, land residual analysis requires that the following conditions be met:

1. The building value of the subject property is known or can be accurately estimated.
2. Net operating income to the property is known or can be estimated.
3. Both building and land capitalization rates can be extracted from the market.

Small variations in any of these variables can result in a dramatic change in the land value indication. (Chapter 30 discusses procedures for estimating building costs, Chapter 24 discusses the development of income and expense estimates, and Chapter 25 discusses the extraction of land and building capitalization rates.)

To apply the land residual technique as an income approach analysis, the appraiser first determines what actual or hypothetical improvements represent the highest and best use of the site as though vacant. Then the net operating income (*NOI* or I_o) of the property is estimated from market rents and operating expenses as of the date of the appraisal. Next, the appraiser calculates how much of the income is attributable to the building and subtracts this amount from the net operating income. The remainder is the residual income attributable to the land, which is capitalized at a market-derived land capitalization rate to provide an estimate of site value.

As a simple example, suppose the subject property's net operating income is \$150,000, the land capitalization rate is 5.5%, the capitalization rate for the building is 7.5%, and the value of the improvements is \$900,000. The calculations of the residual land value would be as follows:

$$\begin{aligned}\text{Net Operating Income } (I_o) &= \$150,000 \\ \text{Income to the Building } (I_b) &= \text{Building Value } (V_b) \times \text{Building Capitalization Rate } (R_b) \\ \text{Income to the Land } (I_L) &= I_o - I_b = \$150,000 - \$67,500 = \$82,500 \\ \text{Site Value} &= \frac{\text{Income to the Land}}{\text{Land Capitalization Rate}} = I_L / R_L = \frac{\$82,500}{0.055} = \$1,500,000\end{aligned}$$

As a cost-based technique, land residual analysis shares characteristics with market extraction in that land value is estimated as the difference between the value of an entire property and the value of the improvements on the property. However, the extraction technique involves analysis of comparable sites, while land residual analysis is focused on the subject property. That is, in the market extraction technique, land values of comparable properties are calculated by subtracting the estimated value of the improvements from the value of the land and improvements, with the resulting indications of land values adjusted and applied to the subject property.

In contrast, in land residual analysis, a similar process is employed, although rather than analyzing comparable properties, the subject property is analyzed on its own. Specifically, the valuation of the project upon completion is estimated using both income capitalization and sales comparison techniques, as appropriate, and those value indications are reconciled. Then, the costs related to constructing the improvements are deducted to arrive at an indication of the land value.

Income Capitalization Techniques

The various income capitalization procedures used to estimate land values rely on information that can be difficult to obtain. Therefore, these techniques are generally not

used as primary valuation techniques except in special situations such as subdivision development analysis. Direct capitalization and yield capitalization techniques are discussed in more detail in Chapters 25, 26, and 27, and examples of land valuation using discounted cash flow analysis are shown in Chapter 28.

Ground Rent Capitalization

Ground rent is the amount paid for the right to use and occupy the land according to the terms of a ground lease. Market-derived capitalization rates are used to convert ground rent into market value. The basic calculations are straightforward, as shown in the following example of a 1-acre plot leased for \$2.25 per square foot with a market-derived land capitalization rate of 5.5%:

$$\begin{aligned}\text{Income to the Land } (I_L) &= \text{Ground Rent} \times \text{Land Area} \\ \text{Income to the Land } (I_L) &= \$2.25 \text{ per sq.ft.} \times 43,560 \text{ sq.ft.} = \$98,010 \\ \text{Site Value} &= \frac{\text{Income to the Land}}{\text{Land Capitalization Rate}} = I_L / R_L = \frac{\$98,000}{0.055} = \$1,782,000\end{aligned}$$

The ground rent capitalization procedure is useful when an analysis of comparable parcels of leased land indicates a range of rents and land capitalization rates. However, it must be recognized that if the capitalization rate used in this analysis is derived from sales of leased properties, the value indication resulting from this analysis is a leased fee value, not a fee simple value. In order for this leased fee value to be useful as an indication of the value of the fee simple interest in the land, an adjustment for property rights may be necessary.

If the current rent and terms of a leased parcel corresponds to market rent and terms for comparable leased parcels with similar highest and best uses then only a property rights adjustment would be necessary. If the ground rent paid under the terms of an existing contract on the subject property does not correspond to the prevailing terms and market rent, the value estimate given the current ground rent must also be adjusted for the difference in these terms to obtain an indication of the market value of the fee simple estate. Ground leases can have different terms and escalation clauses that affect the income stream, so appraisers should consider all the benefits to the lessor during the term of a lease and any option periods and forecast when the reversion of the property will take place.

If the capitalization rate used in the ground rent capitalization analysis is derived from fee simple sales instead of leased fee sales, then the value indication would be for the fee simple, and no property rights adjustment would be required to the indicated value. For example, consider a parcel of land that sold six months ago for \$1 million. Shortly after the sale, the buyer found a tenant for a ground lease of the property at a market rate of \$65,000 per year, and the property was then sold to an investor for \$1.3 million. The indicated capitalization rate for the leased fee interest in the property is 5.0% (\$65,000/\$1,300,000). Application of this capitalization rate to the subject property would result in an indication of the leased fee value. However, this information can also be used to extract an implied capitalization rate for the fee simple interest—6.5% (\$65,000/\$1,000,000). This capitalization rate inherently reflects the time, cost, risk, and entrepreneurial incentive involved in acquiring an unleased parcel of land and getting it leased. In this case, applying the 6.5% capitalization rate to the market rent of the subject property would result in a value indication for the fee simple interest in the property. This example can also be used to illustrate the

magnitude of the property rights adjustment that would be applicable if the sale of a leased fee site was used to assist in valuing a fee simple interest in land. In this case, the indicated property rights adjustment would be -23% ($1 - \$1,000,000/\$1,300,000$).

Subdivision Development Analysis (Discounted Cash Flow Analysis)

Subdivision development analysis may involve tracts of residential, commercial, or industrial land (or a mix of land uses) that are large enough to be subdivided into smaller lots or parcels and sold to builders or end users.² A planned subdivision can create a more intense use of the property when zoning, available utilities, access, and other influential elements are favorably combined.

As a tool in land valuation, subdivision development analysis is primarily used to provide a bulk sale value for a group of subdivision lots, either proposed or existing.³ The technique can also be used to estimate the value of vacant land or finished lots in a proposed subdivision development scenario. Subdivision development analysis relies on discounted cash flow analysis and requires that the property have a highest and best use for development consistent with the proposed development plan in order to reflect the current value of the vacant land. The technique may also be used to test several potential development scenarios in determining the highest and best use of the site as though vacant.

Subdivision development analysis is most useful in developing a current opinion of land value when the highest and best use of the land is for immediate development at the time of the appraisal or when there is evidence of market demand to support financially feasible subdivision development. Market analysis provides the evidence necessary to support absorption estimates, retail proceeds, and other components required to calculate land value using subdivision development analysis.

Valuing finished lots or units in a subdivision is a common assignment for real estate appraisers, but subdivision development analysis is a complex procedure. When there is sufficient, reliable market data, the subdivision development method provides credible results. The technique is most useful for developing the bulk market value of a group of subdivision lots or units, whether existing or proposed. The method uses what is known as a *bulk sale scenario* to develop the value of all the lots to one purchaser. The value indication is most persuasive when the sales comparison method of estimating land value provides additional support.

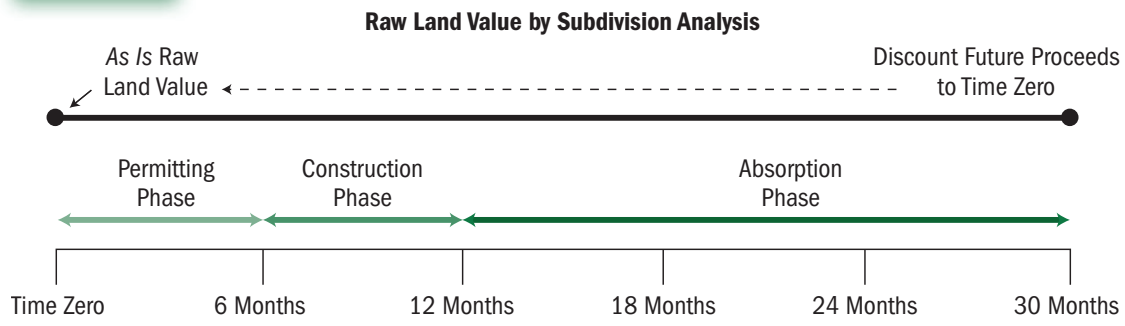
In essence, the subdivision development method uses the yield capitalization techniques of the income capitalization approach to perform a discounted cash flow analysis. To use discounted cash flow analysis to estimate raw land value, appraisers must thoroughly understand the land development process and all the factors influencing the subject property's market area.

The development of any project involves three phases:

- The permitting stage
- The construction stage
- The absorption stage

2. Subdivision development analysis can be applied to completed homes within a subdivision, condominiums, super pads, and other property types, but in this chapter the focus is on finished lots or units within a subdivision.

3. For an in-depth discussion of subdivision development analysis, see Don M. Emerson Jr., *Subdivision Valuation*, 2nd ed. (Chicago: Appraisal Institute, 2017).

Figure 19.1 Sample Income Capitalization Approach Timeline

Source: Don M. Emerson Jr., *Subdivision Valuation*, 2nd ed. (Chicago: Appraisal Institute, 2017), 103.

Data on sales and costs for the finished lots or units must be available. The real estate developer usually provides the necessary project information, including the subdivision plat, the costs of development, a feasibility, marketability, or absorption study, and a schedule of retail lot prices. When the developer supplies the information, an appraiser has a responsibility to compare this information with other relevant market data. The market data and all conclusions in the analysis are the responsibility of the appraiser.

Adequately supported development cost estimates and an adequate investigation of the available market demand needed to support the absorption of lots over time are critical elements of subdivision analysis. In markets that are oversupplied or experiencing an economic downturn, appraisers must not simply assume that market demand will stabilize at some arbitrary point in the future. Future market conditions and demand forecasts must be supported by relevant market data.

To estimate raw land value using the subdivision development technique for a proposed development, an appraiser performs the following steps:

- Assess the highest and best use of the site (e.g., the mix and intensity of uses).
- Create or affirm a supportable subdivision development plan.
- Determine the timing and cost for approval and development (including all costs and conditions of obtaining development entitlements during the permitting stage).
- Forecast a schedule of retail lot prices or values over the absorption stage.
- Critically review available data as necessary to forecast the lot absorption and price mix using inferred or fundamental demand and supply analysis (including properly supported projections of community or market growth over the absorption period).
- Estimate a market-supported timeline for the permitting, construction, and absorption phases.
- Forecast holding costs as well as marketing and related sales expenses over the permitting, construction, and absorption period.
- Estimate the annual real estate taxes and any other miscellaneous expenses over the three stages of development.
- Consider management supervision or administrative costs as part of development expenses.
- Choose an appropriate, market-supported discount rate.

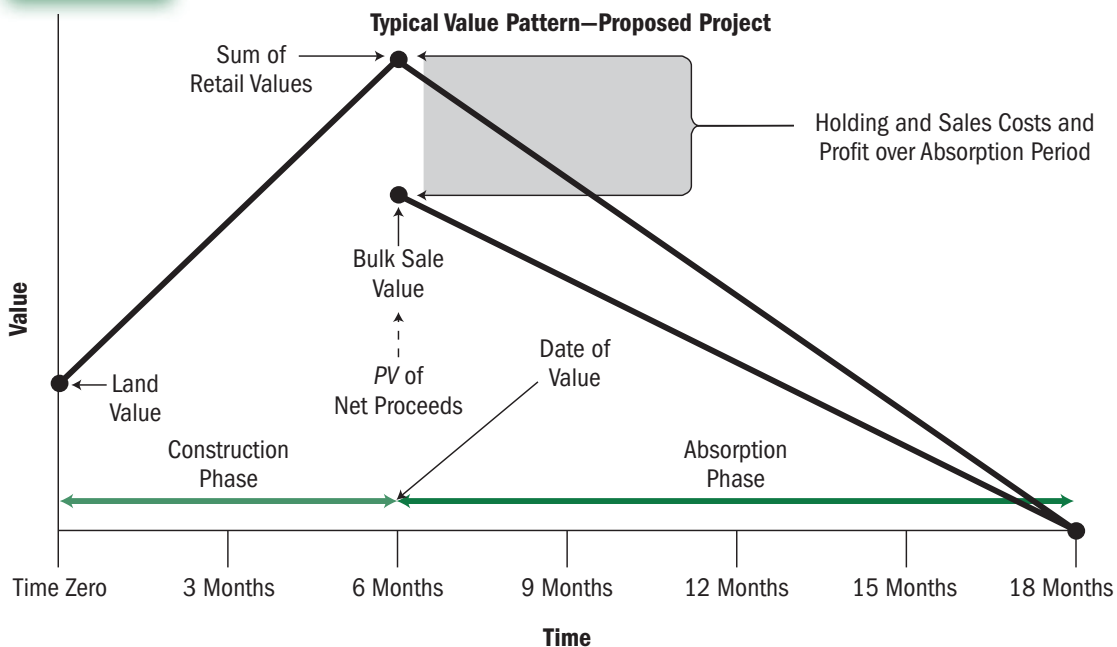
This process can be modified to account for a date of valuation after the preliminary permitting phase of the development process even further along the timeline (see Figure 19.2).

In simplified form, an appraiser begins the analysis of a subdivision development by determining the number and size of the lots that can be created on the appraised parcel of land physically, legally, and economically. The proposed lots must conform to jurisdictional and zoning requirements with regard to size, frontage, topography, soil quality, and off-site improvements (e.g., water facilities, drainage, sewage, streets, curbs, and gutters). The lots must also meet the demands of the market in which the property is located. Without surveys and engineering studies, an appraiser cannot know exactly how many lots can be created from a particular parcel of land. Simply multiplying the allowed density by the site size will not necessarily indicate the number of lots that will be approved on the site, due to setback requirements and other restrictions. The extent of analysis performed may relate to the scope of work of the appraisal assignment. A reasonable estimate of the number of potential lots can often be deduced from zoning information, subdivision ordinances, and the number of lots or typical unit density reflected in similar subdivision developments in the subject's market area. Allowances must also be made to account for the land needed for streets, green space, water retention facilities, and any common areas.

If available, a preliminary development plan for the hypothetical subdivision of the vacant land being appraised will specify much of the data an appraiser needs:

- Number and size of the lots
- Land development or construction work to be accomplished
- Direct (hard) and indirect (soft) construction costs

Figure 19.2 Sample Value Pattern of a Proposed Project



Source: Don M. Emerson Jr., *Subdivision Valuation*, 2nd ed. (Chicago: Appraisal Institute, 2017), 105.

- Probable time required to subdivide the land and construct the on-site and off-site infrastructure
- Expenses and holding costs that will be incurred during the relevant stages of development

The appraiser then undertakes a marketability study to assess the supply and demand situation and the probable capture and absorption rates to produce a supportable absorption forecast for the lot inventory. Well-supported forecasts of product demand and competitive supply can increase the credibility of the absorption forecast. They are critical to the analysis because the raw land value can vary widely depending on the rate at which finished lots will be absorbed over time. The appraiser estimates the projected retail prices of the lots by applying the sales comparison method.

The next step in subdivision development analysis requires a forecast of income and expenses associated with the permitting, construction, and eventual sellout or absorption of the finished lots over time. The time period used for the analysis ends when the last lot is sold. Depending project size, sales velocity, and derivation of the discount rate, annual, semiannual, or quarterly discounting periods are commonly used. The projection period begins with the property in its current condition and, if permitting is required, will include the time frame needed to achieve permitting, construction, and the absorption of all lots. If permitting is already in place as of the date of valuation, then the time period considered in the discounting calculation consists of the construction and absorption phases only. The net cash flows from each period are discounted to time period zero to arrive at a present value of the net proceeds, which is an indication of the value of the property at that point in time.

Banking regulators and examiners like subdivision development analysis because it provides transparency on the inputs used as the basis of the valuation. In economic downturns, the number of sales of bulk finished lots are limited and regulators may require lenders (as appraisal clients) to instruct appraisers as an assignment condition to include a discounted cash flow analysis for the bulk lot holding.